

Dominic Lemoine de Martigny

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Education

University of Waterloo

Bachelor of Software Engineering - Co-op, GPA: 3.88/4.0

Waterloo, ON

Sep. 2024 - Present

Skills

Languages: Python, TypeScript, C, C++, SQL, JavaScript, HTML, CSS, Java, C#, Ruby

Technology: React, Node.js, AWS (Lambda, EC2, S3, CloudFormation, RDS), MySQL, PostgreSQL, Supabase, Firebase, GraphQL, Next.js, Vite, LangChain, Tailwind, Flask, Flutter, PyTorch, OpenCV, Django, FastAPI, Spring Boot, Vue.js

Development: Docker, Git, Kubernetes, Postman, CI/CD, Playwright, GTest, Pytest, Jest, Linux, Colab, Jupyter

Experience

Infrastructure Engineering Intern

Shopify

May 2026 - Aug. 2026

Toronto, ON

- Engineered an **E2E testing framework** for **KateSQL** (MySQL on Kubernetes) using **Ruby on Rails** and **Go** to validate critical database failovers and cross-region recovery, eliminating **300+ manual engineer-hours** annually.
- Built an autonomous validation system on **GitHub Actions** that deploys and tests **Dependabot** PRs in staging, lifting a **150+ PR backlog** from an **11%** to an **80%+** merge rate and saving **\$40K+/year** in engineering time.
- Developed an **LLM-assisted** data pipeline and dashboard to consolidate live performance, schema, and ownership metadata for **178 tables** across **61 teams** to guide a company-wide **MySQL** to **Yugabyte** migration.

Software Engineer Intern

Versaterm

Jan. 2026 - Apr. 2026

Vancouver, BC

- Overhauled **SQL/Hibernate** queries across 3 professional standards products with index optimization and **Redis** caching, reducing dashboard load times by up to **95%** (2 min to 3s) for **1,400+** law enforcement agencies.
- Built **Liquibase** migrations and cleanup scripts across **80+ MS SQL Server** tables to enforce FK constraints and auto-incrementing PKs, eliminating years of legacy orphaned records causing production failures.
- Implemented purview-scoped **Spring Boot** service-layer audit logging to track and filter admin setting changes.
- Refactored a **Vue.js** citizen submission portal to support fully configurable, agency-customizable submission types.

Software Engineer Intern

ALS Geoanalytics

May. 2025 - Aug. 2025

Montréal, QC

- Deployed custom CV segmentation models using **Kubernetes** and **EC2**, cutting labeling time from **~5 min** to **10 s**.
- Developed an **S3** cloud annotation system handling **5TB+** of data, enabling session persistence for geoscientists.
- Engineered a microservices architecture with **FastAPI**, implementing a **RAG**-powered chatbot using **LangChain** and **FAISS** to query thousands of internal research documents, reducing retrieval time by up to **~40%**.
- Built a **React/Django** platform integrating **AWS Lambda** and **RDS** to drive scalable, self-service custom model training, featuring real-time **WebSocket** dashboard updates and **Amplify** authentication.

Projects

uwGuessr.com | TypeScript, Next.js, GraphQL, Supabase, AWS (Lambda, S3), Tailwind, Mapbox, Auth0

- Launched **uwGuessr**, a Waterloo geoguessing web game, reaching **120,000+** visits and **11,000+** unique users.
- Implemented a real-time leaderboard system using **GraphQL** subscriptions and **Supabase** PostgreSQL triggers, processing **200+** daily submissions with sub-**300ms** response times
- Built a user-submitted image upload workflow with client-side compression, **Lambda** processing, and **S3** storage.
- Deployed security features including **rate limiting**, **proxy detection**, and **Auth0** to counter active DDOS attacks.

RememberU (Hack the North) | Flutter, Firebase, Python, Flask, OpenCV, Gemini, Raspberry Pi, Postman

- Built a wearable **AI-powered** device and **Flutter** app with a **Flask + Firebase** backend, combining **lip-reading** and **facial recognition** to capture conversations and deliver personalized memory recall during busy events.
- Integrated **Gemini**, **OpenCV**, and lip-reading APIs to generate real-time, **context-aware** conversation summaries.